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MODULE 2

Environmental Pollution: Air Pollution and its effects, Water pollution and its sources, Zero waste concept and 3 R concepts in solid waste management; Greenhouse effect, Global warming, Climate change, Ozone layer depletion, Carbon credits, carbon trading and carbon foot print, legal provisions for environmental protection.

2.1 AIR POLLUTION

The presence in the atmosphere of any air pollutant means:

“any solid, liquid or gaseous substances (including noise) present in the atmosphere in such concentrations as may be or tend to be injurious to human beings or other living creatures or plants or property or environment”.

Pollution– Presence of foreign substances in the atmosphere in high enough concentration and for long enough duration to cause undesirable effects.

Pollutants – Foreign substances – Particulates, Sulphur dioxide, Nitrogen dioxide, Carbon mono oxide, Hydro carbons, Ozone, Lead

2.2 EFFECTS OF AIR POLLUTION

- Causes Weather changes, temperature inversion. Illness, deaths etc.
- Sulphur dioxide, Nitrogen dioxide, Ozone, cause Eye and throat irritation, coughing and chest paining.
- **Lead fumes** -- asbestos fibers are carcinogenic.
- **Carbon monoxides** -- odorless, goes inside the body and combines with Hemoglobin and absorbs oxygen from the blood causing death
- Due to atmospheric pollution materials also get affected
- Corrosion of Metals
- Deterioration of Textiles, Leather, Rubber, Nylon and other synthetic products.
- Air pollution affects vegetables, animals, fruits, trees, flowers etc.
- Leaves fall, vegetables decay, fruits dry.
- Cattle and livestock die.
- Due to corrosion, the pipe lines of water supply get affected and water system gets choked up
- Due to chemical reaction of the soil the vegetables get decayed or become poisonous
- Water also gets polluted with air pollution

GREEN HOUSE EFFECT

- CO₂ is not an air pollutant in the normal course.
- It presents in air – 0.34 % or 340 ppm.(partspermillion)
- Combustion of coal oil and gas releases CO₂ in atmosphere.
- Less heat will escape out with CO₂ concentration.
- This causes increase in heat level. It is known as Green House Effect

ACID RAIN

- Neutrality of Water is at average 7 pH value.
- pH value of Water ranges between 0 and 14.
- Anything below 7 pH value becomes acidic.
- Anything above 7 pH value becomes alkaline.
- Concentration of SO₂, NO₂ reacts with moisture to form nitric acid and sulphuric acid .This is known as **Acid Rain**.
- This acid rain affects the buildings, other structures and bridges corrode

ENVIRONMENTAL PROBLEMS OF ACID RAIN.

- Contamination of Fresh Water lakes.
- Forest decay.
- Shrubbing of agricultural crops.
- Contamination of drinking water.
- Damage of other materials.
- Death of fish.
- Reduction of soil fertility.
- Deterioration of concrete, metals, materials.

IMPURITIES IN AIR

- SPM – Suspended particulate matter
- CO – Carbon Monoxide
- Other Noxious Gases
- Fly ash from power houses
- Decaying of organic matters
- Municipal waste
- Hazardous chemical waste

2.3 WATER POLLUTION

Water pollution is the contamination of water bodies (e.g. lakes, rivers, oceans, aquifers and groundwater). This occurs when pollutants are directly or indirectly discharged into water bodies without adequate treatment to remove harmful compounds.

Any chemical, biological, or physical change in water quality that has a harmful effect on living organisms or makes water unsuitable for desired usage is known as water pollution.

2.3.1 TYPES OF WATER POLLUTION

There are many types of water pollution because water comes from many sources. Here are a few types of water pollution:

1. Nutrients Pollution

Some wastewater, fertilizers and sewage contain high levels of nutrients. If they end up in water bodies, they encourage algae and weed growth in the water. This will make the water undrinkable, and even clog filters. Too much algae will also use up all the oxygen in the water and other water organisms in the water will die out of oxygen starvation.

2. Surface water pollution

Surface water includes natural water found on the earth's surface, like rivers, lakes, lagoons and oceans. Hazardous substances coming into contact with this surface water, dissolving or mixing physically with the water can be called surface water pollution.

3. Oxygen Depleting

Water bodies have micro-organisms. These include aerobic and anaerobic organisms. When too much biodegradable matter (things that easily decay) end up in water, it encourages more microorganism growth, and they use up more oxygen in the water. If oxygen is depleted, aerobic organisms die, and anaerobic organisms grow more to produce harmful toxins such as ammonia and sulfides.

4. Ground water pollution

When humans apply pesticides and chemicals to soils, they are washed deep into the ground by rain water. This gets to underground water, causing pollution underground. This means when we dig wells and bore holes to get water from underground, it needs to be checked for ground water pollution.

5. Microbiological

In many communities in the world, people drink untreated water (straight from a river or stream). Sometimes there is natural pollution caused by micro-organisms like viruses, bacteria and protozoa. This natural pollution can cause fishes and other water life to die. They can also cause serious illness to humans who drink from such waters.

6. Suspended Matter

Some pollutants (substances, particles and chemicals) do not easily dissolve in water. This kind of material is called particulate matter. Some suspended pollutants later settle under the water body. This can harm and even kill aquatic life that live at the floor of water bodies.

7. Chemical Water Pollution

Many industries and farmers work with chemicals that end up in water. This is common with Point-source Pollution. These include chemicals that are used to control weeds, insects and pests. Metals

and solvents from industries can pollute water bodies. These are poisonous to many forms of aquatic life and may slow their development, make them infertile and kill them.

8. Oil Spillage

Oil spills usually have only a localized effect on wildlife but can spread for miles. The oil can cause the death to many fish and get stuck to the feathers of seabirds causing them to lose their ability to fly.

2.3.2 WATER POLLUTION SOURCES

There are many causes for water pollution but two general categories exist: direct and indirect contaminant sources.

Direct sources include effluent outfalls from factories, refineries, waste treatment plants etc. that emit fluids of varying quality directly into urban water supplies. In the United States and other countries, these practices are regulated, although this doesn't mean that pollutants can't be found in these waters.

Indirect sources include contaminants that enter the water supply from soils/groundwater systems and from the atmosphere via rain water. Soils and ground waters contain the residue of human agricultural practices (fertilizers, pesticides, etc..) and improperly disposed of industrial wastes. Atmospheric contaminants are also derived from human practices (such as gaseous emissions from automobiles, factories and even bakeries).

POINT SOURCES

Point source water pollution refers to contaminants that enter a waterway from a single, identifiable source, such as a pipe or ditch. Examples of sources in this category include discharges from a sewage treatment plant, a factory, or a city storm drain.

NON POINT SOURCES

Nonpoint source pollution refers to diffuse contamination that does not originate from a single discrete source. NPS pollution is often the cumulative effect of small amounts of contaminants gathered from a large area. A common example is the leaching out of nitrogen compounds from fertilized agricultural lands. Nutrient runoffs in storm water from “sheet flow” over an agricultural field or a forest are also cited as examples of NPS pollution.

Contaminated storm water washed off of parking lots, roads and highways, called urban runoff, is sometimes included under the category of NPS pollution. However, because this runoff is typically channeled into storm drain systems and discharged through pipes to local surface waters, it becomes a point source.

2.3.3 CAUSES OF WATER POLLUTION

1. **Industrial waste:** Industries produce huge amount of waste which contains toxic chemicals and pollutants which can cause air pollution and damage to us and our environment. They contain pollutants such as lead, mercury, sulphur, asbestos, nitrates and many other harmful chemicals. Many industries do

not have proper waste management system and drain the waste in the fresh water which goes into rivers, canals and later in to sea. The toxic chemicals have the capability to change the color of water, increase the amount of minerals, also known as Eutrophication, change the temperature of water and pose serious hazard to water organisms.

2. **Sewage and waste water:** The sewage and waste water that is produced by each household is chemically treated and released in to sea with fresh water. The sewage water carries harmful bacteria and chemicals that can cause serious health problems. Pathogens are known as a common water pollutant; The sewers of cities house several pathogens and thereby diseases. Microorganisms in water are known to be causes of some very deadly diseases and become the breeding grounds for other creatures that act like carriers. These carriers inflict these diseases via various forms of contact onto an individual. A very common example of this process would be Malaria.

3. **Mining activities:** Mining is the process of crushing the rock and extracting coal and other minerals from underground. These elements when extracted in the raw form contain harmful chemicals and can increase the amount of toxic elements when mixed up with water which may result in health problems. Mining activities emit several metal waste and sulphides from the rocks and is harmful for the water.

4. **Marine dumping:** The garbage produce by each household in the form of paper, aluminum, rubber, glass, plastic, food if collected and deposited into the sea in some countries. These items take from 2 weeks to 200 years to decompose. When such items enter the sea, they not only cause water pollution but also harm animals in the sea.

5. **Accidental Oil leakage:** Oil spill pose a huge concern as large amount of oil enters into the sea and does not dissolve with water; there by opens problem for local marine wildlife such as fish, birds and sea otters. For e.g.: a ship carrying large quantity of oil may spill oil if met with an accident and can cause varying damage to species in the ocean depending on the quantity of oil spill, size of ocean, toxicity of pollutant

6. **Burning of fossil fuels:** Fossil fuels like coal and oil when burnt produce substantial amount of ash in the atmosphere. The particles which contain toxic chemicals when mixed with water vapor result in acid rain. Also, carbon dioxide is released from burning of fossil fuels which result in global warming.

7. **Chemical fertilizers and pesticides:** Chemical fertilizers and pesticides are used by farmers to protect crops from insects and bacteria. They are useful for the plants growth. However, when these chemicals are mixed up with water produce harmful for plants and animals. Also, when it rains, the chemicals mixes up with rainwater and flow down into rivers and canals which pose serious damages for aquatic animals.

8. **Leakage from sewer lines:** A small leakage from the sewer lines can contaminate the underground water and make it unfit for the people to drink. Also, when not repaired on time, the leaking water can come on to the surface and become a breeding ground for insects and mosquitoes.

9. **Global warming:** An increase in earth's temperature due to greenhouse effect results in global warming. It increases the water temperature and result in death of aquatic animals and marine species which later results in water pollution.

10. **Radioactive waste:** Nuclear energy is produced using nuclear fission or fusion. The element that is used in production of nuclear energy is Uranium which is highly toxic chemical. The nuclear waste that is produced by radioactive material needs to be disposed off to prevent any nuclear accident. Nuclear waste can have serious environmental hazards if not disposed off properly. Few major accidents have already taken place in Russia and Japan.

11. **Urban development:** As population has grown, so has the demand for housing, food and cloth. As more cities and towns are developed, they have resulted in increase use of fertilizers to produce more food, soil erosion due to deforestation, increase in construction activities, inadequate sewer collection and treatment, landfills as more garbage is produced, increase in chemicals from industries to produce more materials.

12. **Leakage from the landfills:** Landfills are nothing but huge pile of garbage that produces awful smell and can be seen across the city. When it rains, the landfills may leak and the leaking landfills can pollute the underground water with large variety of contaminants.

13. **Animal waste:** The waste produce by animals is washed away into the rivers when it rains. It gets mixed up with other harmful chemicals and causes various water borne diseases like cholera, diarrhea, jaundice, dysentery and typhoid.

14. **Underground storage leakage:** Transportation of coal and other petroleum products through underground pipes is well known. Accidentals leakage may happen anytime and may cause damage to environment and result in soil erosion.

Water pollutants also include both organic and inorganic factors. Organic factors include volatile organic compounds, fuels, waste from trees, plants etc. Inorganic factors include ammonia, chemical waste from factories, discarded cosmetics etc. The water that travels via fields is usually contaminated with all forms of waste inclusive of fertilizers that it swept along the way. This infected water makes its way to our water bodies and sometimes to the seas endangering the flora, fauna and humans that use it along its path.

The current scenario has led to a consciousness about water preservation and efforts are being made on several levels to redeem our water resources. Industries and factory set-up's are restricted from contaminating the water bodies and are advised to treat their contaminated waste through filtration methods. People are investing in rain water harvesting projects to collect rainwater and preserve it in wells below ground level.

2.3.3 EFFECTS OF WATER POLLUTION

The effects of water pollution are varied and depend on what chemicals are dumped and in which locations. Many water bodies near urban areas (cities and towns) are highly polluted. This is the result of both garbage dumped by individuals and dangerous chemicals legally or illegally dumped by manufacturing industries, health centers, schools and market places.

1. Death of aquatic (water) animals

The main problem caused by water pollution is that it kills life that depends on these water bodies. Dead fish, crabs, birds and sea gulls, dolphins, and many other animals often wind up on beaches, killed by pollutants in their habitat (living environment).

2. Disruption of food-chains

Pollution disrupts the natural food chain as well. Pollutants such as lead and cadmium are eaten by tiny animals. Later, these animals are consumed by fish and shellfish, and the food chain continues to be disrupted at all higher levels

3. Diseases

Eventually, humans are affected by this process as well. People can get diseases such as hepatitis by eating seafood that has been poisoned. In many poor nations, there is always outbreak of cholera and diseases as a result of poor drinking water treatment from contaminated waters.

4. Destruction of ecosystems

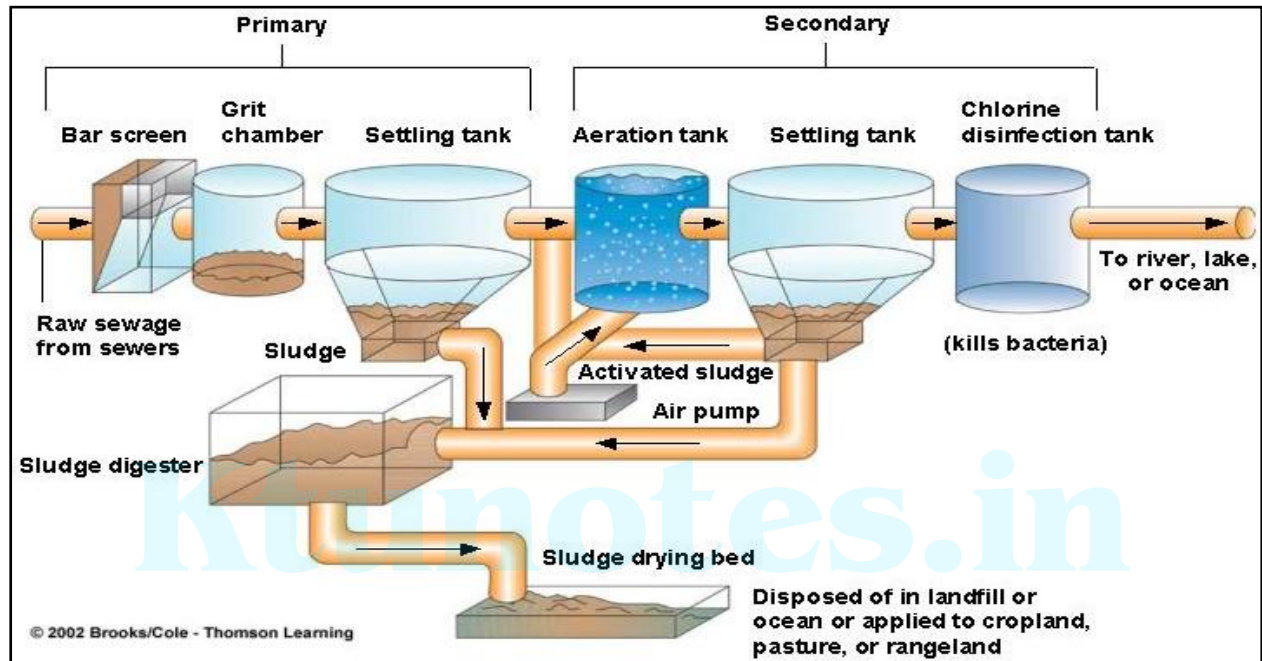
Ecosystems (the interaction of living things in a place, depending on each other for life) can be severely changed or destroyed by water pollution. Many areas are now being affected by careless human pollution, and this pollution is coming back to hurt humans in many ways.

2.4 SUSTAINABLE WASTE WATER TREATMENT

- ❖ Groundwater Pollution: Causes
 - Low flow rates
 - Few bacteria
 - Low oxygen
 - Cold temperatures
- ❖ Groundwater Pollution Prevention
 - Monitor aquifers
 - Find less hazardous substitutes
 - Leak detection systems
 - Strictly regulating hazardous waste disposal
 - Store hazardous materials above ground
- ❖ Preventing and Reducing Surface Water Pollution
- ❖ Nonpoint Sources
 - Reduce runoff
 - Buffer zone vegetation
 - Reduce soil erosion
- ❖ Point Sources
 - Clean Water Act
 - Water Quality Act
 - Clean Water Act
 - Sets standards for key pollutants

- Requires permits for discharge
- Requires sewage treatment
- Require permits for wetland destruction
- Does not deal with nonpoint sources well
- Goal All Waterways fishable and swimmable

2.4.1 SEWAGE TREATMENT



2.4.2 LEVELS OF TREATMENT

Primary

Removal by physical separation of grit and large objects (material to landfill for disposal)

Secondary

Aerobic microbiological process (sludge)

Organic matter + O₂ - CO₂ + NH₃ + H₂O

NH₃ -NO₃-

Lowers suspended solids content (into sludge)

Tertiary (advanced)

Anaerobic microbiological process with a different microbe where O₂ is toxic (more sludge)

2.4.3 Liquid Waste (Sewage/Wastewater) Treatment

Wastewater (liquid waste) from flushing the toilet, bathing, washing sinks and general cleaning goes down the drain and into a pipe, which joins a larger sewer pipe under the road. The larger pipe also joins a major pipe that leads to the treatment center.

STAGE ONE: SCREENING

Screening is first stage of the wastewater treatment process. Screening removes large objects like, diapers, nappies, sanitary items, cotton buds, face wipes and even broken bottles, bottle tops, plastics and rags that may block or damage equipment.

Special equipment is also used to remove grit that gets washed into the sewer.

STAGE TWO: PRIMARY TREATMENT

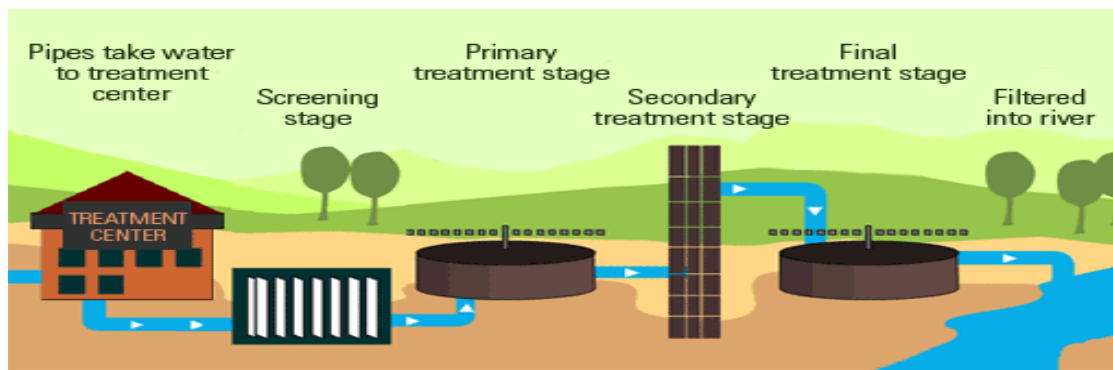
This involves the separation of organic solid matter (or human waste) from the wastewater. This is done by putting the wastewater into large settlement tanks for the solids to sink to the bottom of the tank. The settled solids are called 'sludge'. At the bottom of these circular tanks, large scrapers continuously scrape the floor of the tank and push the sludge towards the center where it is pumped away for further treatment. The rest of the water is then moved to the Secondary treatment.

STAGE THREE: SECONDARY TREATMENT

The water, at this stage is put into large rectangular tanks. These are called aeration lanes. Air is pumped into the water to encourage bacteria to breakdown the tiny bits of sludge that escaped the sludge scraping process.

STAGE FOUR: FINAL TREATMENT

Next the 'almost' treated wastewater is passed through a settlement tank. Here, more sludge is formed at the bottom of the tank from the settling of the bacterial action. Again, the sludge is scrapped and collected for treatment. The water at this stage is almost free from harmless substances and chemicals. The water is allowed to flow over a wall where it is filtered through a bed of sand to remove any additional particles. The filtered water is then released into the river.



2.5 WASTE WATER TREATMENT

Are also classified in to

- Physical treatment
- Chemical treatment
- Biological treatment

The removal of wastes by using physical forces is:

Screening, mixing, flocculation, sedimentation, floatation, heat transfer and drying

In chemical treatment the pollutants are removed by adding chemicals, this method is not used in the treatment of domestic water

In biological process the pollutants are removed by biological activity of microorganisms

Activated sludge process

Aerated lagoon

Oxidation pond

Aerobic and anaerobic digesters

Trickling filters

2.5.1 SUSTAINABLE TREATMENT TYPES

- Lagoons/wetlands
- Anaerobic digestion
- Soil aquifer treatment

In wetland treatment natural forces (physical, chemical, solar) act together to purify the waste water, thereby achieving waste water treatment. A series of shallow ponds act as stabilization lagoons, while water hyacinth or duckweed acts to accumulate heavy metals. multiple forms of bacteria , plankton and algae act to further purify the water

Advantages

- It is mechanized treatment systems
- Allows for total resource recovery
- Low cost

Disadvantages

- Affected by climate change
- Some locations may make it unsustainable
- Mechanical problems may include clogging with sprinkler and drip irrigation systems
- High demand for flat land
- Effluents contain valuable fertilizers

Soil aquifer treatment is a geo purification system where partially treated sewage effluent artificially recharges the aquifer and then withdrawn for future use. By recharging through unsaturated soil layers, the effluent achieves additional purification before it is mixed with the natural ground water.

Advantages

- Cost is lower
- Remove pathogens efficiently
- Operation is not highly technical

Disadvantages

- Can change the hydrological properties of ground water and soil
- Requires big area for infiltration

2.6 SOLID WASTE MANAGEMENT

Any material that we discard, that is not liquid or gas, is solid waste

- Municipal Solid Waste (MSW):
- Solid waste from home or office
- Industrial Solid Waste:
- Solid waste produced from Mines, Agriculture or Industry

Types

Hazardous waste is a waste with properties that make it dangerous or potentially harmful to human health or the environment

Ignitability - Ignitable wastes create fires under certain conditions or are spontaneously combustible, or have a flash point less than 60 °C (140 °F).

Corrosivity - Corrosive wastes are acids or bases (pH less than or equal to 2 or greater than or equal to 12.5) that are capable of corroding metal containers, such as storage tanks, drums, and barrels.

Reactivity - Reactive wastes are unstable under "normal" conditions. They can cause explosions, toxic fumes, gases, or vapors when mixed with water.

Toxicity - Toxic wastes are harmful or fatal when ingested or absorbed (e.g., containing mercury, lead, etc.). When toxic wastes are disposed of on land, contaminated liquid may drain (leach) from the waste and pollute ground water. Toxicity is defined through a laboratory procedure called the Toxicity Characteristic Leaching Procedure (TCLP).

2.6.1 CLASSIFICATION OF WASTES ACCORDING TO THEIR PROPERTIES

- **Bio-degradable**
can be degraded (paper, wood, fruits and others)
- **Non-biodegradable**
Cannot be degraded (plastics, bottles, old machines, cans, Styrofoam containers and others)

2.6.2 CLASSIFICATION OF WASTES ACCORDING TO THEIR EFFECTS ON HUMAN HEALTH AND THE ENVIRONMENT

- **Hazardous wastes**
Substances unsafe to use commercially, industrially, agriculturally, or economically and have any of the following properties- ignitability, corrosivity, reactivity & toxicity.
- **Non-hazardous**
Substances safe to use commercially, industrially, agriculturally, or economically and do not have any of those properties mentioned above. These substances usually create disposal problems.

2.6.3 CLASSIFICATION OF WASTES ACCORDING TO THEIR ORIGIN AND TYPE

Municipal Solid wastes: Solid wastes that include household garbage, rubbish, construction & demolition debris, sanitation residues, packaging materials, trade wastes etc. are managed by any municipality.

Bio-medical wastes: Solid or liquid wastes including containers, intermediate or end products generated during diagnosis, treatment & research activities of medical sciences.

Industrial wastes: Liquid and solid wastes that are generated by manufacturing & processing units of various industries like chemical, petroleum, coal, metal gas, sanitary & paper etc.

Agricultural wastes: Wastes generated from farming activities. These substances are mostly biodegradable.

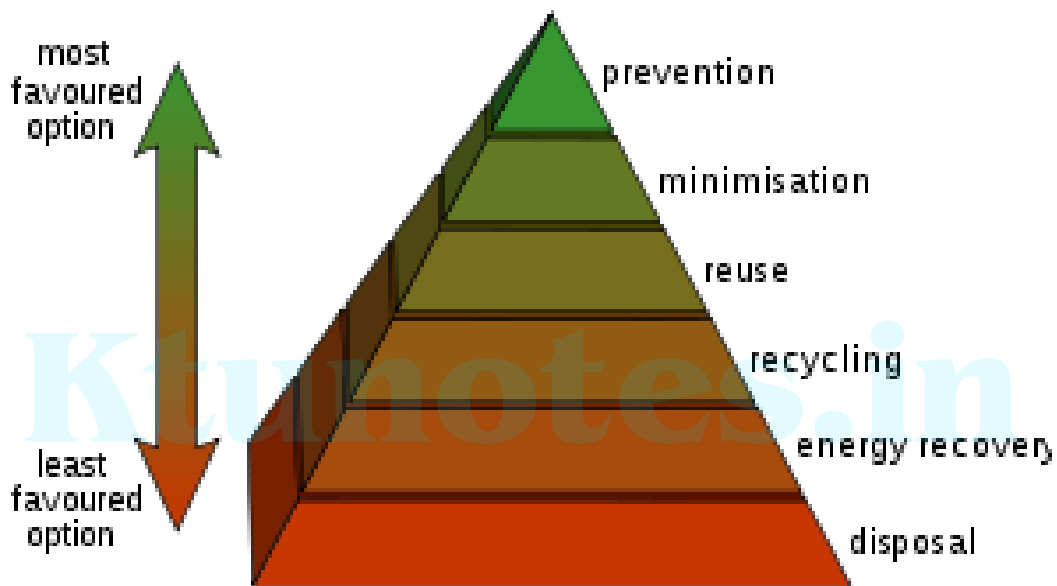
Fishery wastes: Wastes generated due to fishery activities. These are extensively found in coastal & estuarine areas.

Radioactive wastes: Waste containing radioactive materials. Usually these are byproducts of nuclear processes. Sometimes industries that are not directly involved in nuclear activities, may also produce some radioactive wastes, e.g. radio-isotopes, chemical sludge etc.

E-wastes: Electronic wastes generated from any modern establishments. They may be described as discarded electrical or electronic devices. Some electronic scrap components, such as CRTs, may contain contaminants such as Pb, Cd, Be or brominated flame retardants.

2.6.4 METHODS OF WASTE DISPOSAL

- Landfills
- Incineration
- Source reduction
- Composting
- Recycling
- reusing



2.6.5 IMPACTS OF WASTE ON HEALTH

- Chemical poisoning through chemical inhalation
- Uncollected waste can obstruct the storm water runoff resulting in flood
- Low birth weight
- Cancer
- Congenital malformations
- Neurological disease
- Nausea and vomiting
- Increase in hospitalization of diabetic residents living near hazard waste sites.
- Mercury toxicity from eating fish with high levels of mercury

Effects of waste on animals and aquatics life

- Increase in mercury level in fish due to disposal of mercury in the rivers.
- Plastic found in oceans ingested by birds.
- Resulted in high algal population in rivers and sea.
- Degrades water and soil quality.

Impacts of waste on Environment

- Waste breaks down in landfills to form methane, a potent greenhouse gas
- Change in climate and destruction of ozone layer due to waste biodegradable
- Littering, due to waste pollutions, illegal dumping, Leaching: is a process by which solid waste enter soil and ground water and contaminating them.

2.7 ZERO WASTE CONCEPTS

Zero Waste means designing and managing products and processes to systematically avoid and eliminate the volume and toxicity of waste and materials, conserve and recover all resources, and not burn or bury them.

"Zero Waste is a goal that is ethical, economical, efficient and visionary, to guide people in changing their lifestyles and practices to emulate sustainable natural cycles, where all discarded materials are designed to become resources for others to use.

2.8 GLOBAL ENVIRONMENTAL ISSUES

2.8.1 RESOURCE DEGRADATION

Environmental degradation is the deterioration of the environment through depletion of resources such as air, water and soil; the destruction of ecosystems and the extinction of wildlife

2.8.2 WATER DETERIORATION

One major component of environmental degradation is the depletion of the resource of fresh water on Earth. Approximately only 2.5% of all of the water on Earth is fresh water, with the rest being salt water. 69% of the fresh water is frozen in ice caps located on Antarctica and Greenland, so only 30% of the 2.5% of fresh water is available for consumption. Fresh water is an exceptionally important resource, since life on Earth is ultimately dependent on it. Water transports nutrients and chemicals within the biosphere to all forms of life, sustain both plants and animals, and mould the surface of the Earth with transportation and deposition of materials.

The current top three uses of fresh water account for 95% of its consumption;

- Approximately 85% is used **for irrigation of farmland**
- 6% is used for **domestic purposes** such as indoor bathing uses and outdoor garden and
- 4% is used for **industrial purposes** such as processing, washing, and cooling in

manufacturing centers. It is estimated that one in three people over the entire globe are already facing water shortages, almost one-fifth of the world's population live in areas of physical water scarcity, and almost one quarter of the world's population live in a developing country that lacks the necessary infrastructure to use water from available rivers and aquifers. Water scarcity is an increasing problem due to population growth, increased urbanization, higher standards of living, and climate change.

2.8.3 CLIMATE CHANGE AND TEMPERATURE

Climate change affects the Earth's water supply in a large number of ways. It is predicted that the mean global temperature will rise in the coming years due to a number of forces affecting the climate, the amount of atmospheric **CO₂** will rise, and both of these will influence water resources; evaporation depends strongly on temperature and moisture availability, which can ultimately affect the amount of water available to replenish groundwater supplies.

Transpiration from plants can be affected by a rise in atmospheric CO₂, which can decrease their use of water, but can also raise their use of water from possible increases of leaf area. Temperature increase can decrease the length of the snow season in the winter and increase the intensity of snowmelt in warmer seasons, leading to peak runoff of snowmelt earlier in the season, affecting soil moisture, flood and drought risks, and storage capacities depending on the area.

Warmer winter temperatures cause a decrease in snowpack, which can result in diminished water resources during summer. This is especially important at mid-latitudes and in mountain regions that depend on glacial runoff to replenish their river systems and groundwater supplies, making these areas increasingly vulnerable to water shortages over time; an increase in temperature will initially result in a rapid rise in water melting from glaciers in the summer, followed by a retreat in glaciers and a decrease in the melt and consequently the water supply every year as the size of these glaciers get smaller and smaller.

Thermal expansion of water and increased melting of oceanic glaciers from an increase in temperature gives way to a rise in sea level, which can affect the fresh water supply of coastal areas as

well; as river mouths and deltas with higher salinity get pushed further inland, an intrusion of saltwater results in an increase of salinity in reservoirs and aquifers. Sea-level rise may also consequently be caused by a depletion of groundwater, as climate change can affect the hydrologic cycle in a number of ways. Uneven distributions of increased temperatures and increased precipitation around the globe results in water surpluses and deficits, but a global decrease in groundwater suggests a rise in sea level, even after melt water and thermal expansion were accounted for, which can provide a positive feedback to the problems sea-level rise causes to fresh-water supply.

A rise in air temperature results in a rise in water temperature, which is also very significant in water degradation, as the water would become more susceptible to bacterial growth. An increase in water temperature can also affect ecosystems greatly because of a species' sensitivity to temperature, and also by inducing changes in a body of water's self-purification system from decreased amounts of dissolved oxygen in the water due to rises in temperature.

2.8.4 CLIMATE CHANGE AND PRECIPITATION

A rise in global temperatures is also predicted to correlate with an increase in global precipitation, but because of increased runoff, floods, increased rates of soil erosion, and mass movement of land, a decline in water quality is probable, while water will carry more nutrients, it will also carry more contaminants. While most of the attention about climate change is directed towards global warming and greenhouse effect, some of the most severe effects of climate change are likely to be from changes in precipitation, runoff, and soil moisture. It is generally expected that, on average, global precipitation will increase, with some areas receiving increases and some decreases.

Climate models show that while some regions should expect an increase in precipitation, such as in the tropics and higher latitudes, other areas are expected to see a decrease, such as in the subtropics; this will ultimately cause a latitudinal variation in water distribution. The areas receiving more precipitation are also expected to receive this increase during their winter and actually become drier during their summer, creating even more of a variation of precipitation distribution. Naturally, the distribution of precipitation across the planet is very uneven, causing constant variations in water availability in respective locations.

Changes in precipitation affect the timing and magnitude of floods and droughts, shift runoff processes, and alter groundwater recharge rates. Vegetation patterns and growth rates will be directly

affected by shifts in precipitation amount and distribution, which will in turn affect agriculture as well as natural ecosystems. Decreased precipitation will deprive areas of water, causing water tables to fall and reservoirs and wetlands, rivers, and lakes to empty, and possibly an increase in evaporation and transpiration, depending on the accompanied rise in temperature. Groundwater reserves will be depleted, and the remaining water has a greater chance of being of poor quality from saline or contaminants on the land surface.

In areas with decreasing precipitation as predicted by the climate models, soil moisture may be substantially reduced. With this in mind, agriculture in most areas needs irrigation already, which depletes fresh water supplies both by the physical use of the water and the degradation agriculture causes to the water. Irrigation increases salt and nutrient content in areas that would not normally be affected, and damages streams and rivers from damming and removal of water. Fertilizer enters both human and livestock waste streams that eventually enter groundwater, while nitrogen, phosphorus, and other chemicals from fertilizer can acidify both soils and water. Certain agricultural demands may increase more than others with an increasingly wealthier global population, and meat is one commodity expected to double global food demand by 2050, which directly affects the global supply of fresh water. Cows need water to drink, more if the temperature is high and humidity is low, and more if the production system the cow is in is extensive, since finding food takes more effort. Water is needed in processing of the meat, and also in the production of feed for the livestock. Manure can contaminate bodies of freshwater, and slaughterhouses, depending on how well they are managed, contribute waste such as blood, fat, hair, and other bodily contents to supplies of fresh water.

The transfer of water from agricultural to urban and suburban use raises concerns about agricultural sustainability, rural socioeconomic decline, food security, an increased carbon footprint from imported food, and decreased foreign trade balance. The depletion of fresh water, as applied to more specific and populated areas, increases fresh water scarcity among the population and also makes populations susceptible to economic, social, and political conflict in a number of ways; rising sea levels forces migration from coastal areas to other areas farther inland, pushing populations closer together breaching borders and other geographical patterns, and agricultural surpluses and deficits from the availability of water induce trade problems and economies of certain areas. Climate change is an important cause of involuntary migration and forced displacement.

2.8.5 CAUSES OF ENVIRONMENTAL DEGRADATION

Some environmental life species require substantial areas to help provide food, living space, and other different assets. These creatures are called area specific. It gets to be more troublesome for the wildlife to get the assets they need in order to survive.

1. **Land Disturbance:** A more basic cause of environmental degradation is land damage. Numerous weedy plant species, for example, garlic mustard, are both foreign and obtrusive. A rupture in the environmental surroundings provides for them a chance to start growing and spreading. These plants can assume control over nature, eliminating the local greenery. The result is territory with a solitary predominant plant which does not give satisfactory food assets to all the environmental life. Whole environments can be destroyed because of these invasive species.
2. **Pollution:** Pollution, in whatever form, whether it is air, water, land or noise is harmful for the environment. Air pollution causes health issues. Water pollution degrades the quality of water that we use for drinking purposes. Land pollution results in degradation of earth's surface as a result of human activities. Noise pollution can cause irreparable damage to our ears when exposed to continuous large sounds like honking of vehicles on a busy road or machines producing large noise in a factory or a mill.
3. **Overpopulation:** Rapid population growth puts strain on natural resources which results in degradation of our environment. Mortality rate has gone down due to better medical facilities which have resulted in increased lifespan. More population simple means more demand for food, clothes and shelter. You need more space to grow food and provide homes to millions of people. This results in deforestation which is another factor of environmental degradation.
4. **Landfills:** Landfills pollute the environment and destroy the beauty of the city. Landfills come within the city due the large amount of waste that gets generated by households, industries, factories and hospitals. Landfills pose a great risk to the health of the environment and the people who live there. Landfills produce foul smell when burned and cause huge environmental degradation.
5. **Deforestation:** Deforestation is the cutting down of trees to make way for more homes and industries. Rapid growth in population and urban sprawl are two of the major causes of deforestation. Apart from that, use of forest land for agriculture, animal grazing, harvest for fuel wood and logging is some of the other causes of deforestation. Deforestation contributes to global warming as decreased forest size puts carbon back into the environment.
6. **Natural Causes:** Things like avalanches, quakes, tidal waves, storms, and wildfires can totally crush nearby animal and plant groups to the point where they can no longer survive in those

areas. This can either come to fruition through physical demolition as the result of a specific disaster, or by the long term degradation of assets by the presentation of an obtrusive foreign species to the environment. The latter frequently happens after tidal waves, when reptiles and bugs are washed ashore.

Earth itself causes ecological issues, as well. While environmental degradation is most normally connected with the things that people do, the truth of the matter is that the environment is always changing. With or without the effect of human exercises, a few biological systems degrade to the point where they can't help the life that is supposed to live there.

2.8.6 EFFECTS OF ENVIRONMENTAL DEGRADATION

1. **Impact on Human Health:** Human health might be at the receiving end as a result of the environmental degradation. Areas exposed to toxic air pollutants can cause respiratory problems like pneumonia and asthma. Millions of people are known to have died of due to indirect effects of air pollution.
2. **Loss of Biodiversity:** Biodiversity is important for maintaining balance of the ecosystem in the form of combating pollution, restoring nutrients, protecting water sources and stabilizing climate. **Deforestation, global warming, overpopulation and pollution** are few of the major causes for loss of biodiversity
3. **Ozone Layer Depletion:** Ozone layer is responsible for protecting earth from harmful ultraviolet rays. The presence of **chlorofluorocarbons**, hydro chlorofluorocarbons in the atmosphere is causing the ozone layer to deplete. As it will deplete, it will emit harmful radiations back to the earth.
4. **Loss for Tourism Industry:** The deterioration of environment can be a huge setback for tourism industry that rely on tourists for their daily livelihood. Environmental damage in the form of loss of green cover, loss of biodiversity, huge landfills, increased air and water pollution can be a big turn off for most of the tourists.
5. **Economic Impact:** The huge cost that a country may have to borne due to environmental degradation can have big economic impact in terms of restoration of green cover, cleaning up of landfills and protection of endangered species. The economic impact can also be in terms of loss of tourism industry.

As you can see, there are a lot of things that can have an effect on the environment. If we are not careful, we can contribute to the environmental degradation that is occurring all around the world. We can, however, take action to stop it and take care of the world that we live in by providing environmental education to the people which will help them pick familiarity with their surroundings that will enable to take care of environmental concerns thus making it more useful and protected for our children and other future generations.

2.9 OZONE LAYER DEPLETION

The ozone layer is a belt of naturally occurring ozone gas .It located at 9.3 to 18.6 miles (15 to 30 kilometers) above Earth and serves as a shield from the harmful ultraviolet B radiation emitted by the sun. Ozone is a highly reactive molecule that contains three oxygen atoms. It is constantly being formed and broken down in the high atmosphere, 6.2 to 31 miles (10 to 50 kilometers) above Earth, in the region called the stratosphere.

The ozone layer is deteriorating due to the release of pollution containing the chemicals **chlorine** and **bromine**. This deterioration allows large amounts of ultraviolet B rays to reach Earth, which can cause skin cancer and cataracts in humans and harm animals as well.

Extra ultraviolet B radiation reaching Earth also inhibits the reproductive cycle of phytoplankton, single-celled organisms such as algae that make up the bottom rung of the food chain. The reductions in phytoplankton populations will lower the populations of other animals. Researchers also have documented changes in the reproductive rates of young fish, shrimp, and crabs as well as frogs and salamanders exposed to excess ultraviolet B.

Chlorofluorocarbons (CFCs), chemicals found mainly in spray aerosols heavily used by industrialized nations are the primary culprits in ozone layer breakdown. When CFCs reach the upper atmosphere, they are exposed to ultraviolet rays, which cause them to break down into substances that include chlorine. The chlorine reacts with the oxygen atoms in ozone and rips apart the ozone molecule.

One atom of chlorine can destroy more than a hundred thousand ozone molecules,

The ozone layer above the Antarctic has been particularly impacted by pollution. This regions low temperatures speed up the conversion of CFCs to chlorine. In the southern spring and summer, when the sun shines for long periods of the day, chlorine reacts with ultraviolet rays, destroying ozone on a massive scale, up to 65 percent. This is referred to as the "**ozone hole**."

2.9.1 HEALTH AND ENVIRONMENTAL EFFECTS OF OZONE LAYER DEPLETION

Reductions in stratospheric ozone levels will lead to higher levels of UVB reaching the Earth's surface. The sun's output of UVB does not change; rather, less ozone means less protection, and hence more UVB reaches the Earth. Studies have shown that in the Antarctic, the amount of UVB measured at the surface can double during the annual ozone hole. Another study confirmed the relationship between reduced ozone and increased UVB levels in Canada during the past several years.

2.9.1.1 EFFECTS ON HUMAN HEALTH

Laboratory and epidemiological studies demonstrate that UVB causes melanoma, skin cancer and plays a major role in malignant melanoma development. In addition, UVB has been linked to cataracts -- a clouding of the eye's lens. All sunlight contains some UVB, even with normal stratospheric ozone levels. It is always important to protect your skin and eyes from the sun. Ozone layer depletion increases the amount of UVB and the risk of health effects.

2.9.1.2 EFFECTS ON PLANTS

Physiological and developmental processes of plants are affected by UVB radiation, even by the amount of UVB in present-day sunlight.

Indirect changes caused by UVB are

- Changes in plant form,
- How nutrients are distributed within the plant,
- Timing of developmental phases and
- Secondary metabolism)

may be equally, or sometimes more, important than damaging effects of UVB. These changes can have important implications for plant competitive balance, herbivory, plant diseases, and biogeochemical cycles.

2.9.1.3 EFFECTS ON MARINE ECOSYSTEMS

➤ The increase in the UVB decreases the production of phytoplankton

Phytoplankton forms the foundation of aquatic food webs. Phytoplankton productivity is limited to the euphotic zone, the upper layer of the water column in which there is sufficient sunlight to support net productivity. The position of the organisms in the

euphotic zone is influenced by the action of wind and waves. In addition, many phytoplankton are capable of active movements that enhance their productivity and, therefore, their survival. Exposure to solar UVB radiation has been shown to affect both orientation mechanisms and motility in phytoplankton, resulting in reduced survival rates for these organisms..

- Solar UVB radiation causes **damage to early developmental stages of fish, shrimp, crab**, amphibians and other animals.
- The most severe effects are **decreased reproductive capacity** and impaired larval development.
- Small increases in UVB exposure could result in significant reduction in the size of the population of animals that eat these smaller creatures.

2.9.1.4 EFFECTS ON BIOGEOCHEMICAL CYCLES

Increases in solar UV radiation could affect terrestrial and aquatic biogeochemical cycles, thus altering both sources and sinks of greenhouse and chemically-important trace gases e.g., carbon dioxide (CO₂), carbon monoxide (CO), carbonyl sulfide (COS) and possibly other gases, including ozone. These potential changes would contribute to biosphere-atmosphere feedbacks that attenuate or reinforce the atmospheric buildup of these gases.

2.9.1.5 EFFECTS ON MATERIALS

Synthetic polymers, naturally occurring biopolymers, as well as some other materials of commercial interest are adversely affected by solar UV radiation. These materials can be protected from UVB by special additives. Therefore, **any increase in solar UVB levels will therefore accelerate their breakdown, limiting the length of time** for which they are useful outdoors.

2.10 CARBON CREDITS

A carbon credit is a tradable certificate that allows the holder to emit one tonne of carbon dioxide into the atmosphere above the current emissions quota. Carbon credit systems are a result of both national and international efforts to halt the steady growth in greenhouse gases (GHGs) being dumped into the atmosphere

A carbon credit is a generic term for any tradable certificate or permit representing the right to emit one tonne of carbon dioxide or the mass of another greenhouse gas with a carbon dioxide equivalent (tCO₂e) equivalent to one tonne of carbon dioxide.

Carbon credits and carbon markets are a component of national and international attempts to mitigate the growth in concentrations of greenhouse gases (GHGs).

One carbon credit is equal to one tonne of carbon dioxide, or

In some markets, carbon dioxide equivalent gases. Carbon trading is an application of an emissions trading approach. Greenhouse gas emissions are capped and then markets are used to allocate the emissions among the group of regulated sources.

The goal is to allow market mechanisms to drive industrial and commercial processes in the direction of low emissions or less carbon intensive approaches than those used when there is no cost to emitting carbon dioxide and other GHGs into the atmosphere. Since GHG mitigation projects generate credits, this approach can be used to finance carbon reduction schemes between trading partners and around the world.

There are also many companies that sell carbon credits to commercial and individual customers who are interested in lowering their carbon footprint on a voluntary basis. These carbon offsetters purchase the credits from an investment fund or a carbon development company that has aggregated the credits from individual projects. Buyers and sellers can also use an exchange platform to trade, such as the Carbon Trade Exchange, which is like a stock exchange for carbon credits. The quality of the credits is based in part on the validation process and sophistication of the fund or development company that acted as the sponsor to the carbon project. This is reflected in their price; voluntary units typically have less value than the units sold through the rigorously validated Clean Development Mechanism.

2.10.1 HOW BUYING CARBON CREDITS CAN REDUCE EMISSIONS

Carbon credits create a market for reducing greenhouse emissions.

For example, consider a business that owns a factory putting out 100,000 tonnes of greenhouse gas emissions in a year. Its government enacts a law to limit the emissions that the business can produce. So the factory is given a quota of say 80,000 tonnes per year. The factory either reduces its emissions to 80,000 tonnes or is required to purchase carbon credits to offset the excess. After costing up alternatives the business may decide that it is uneconomical or infeasible to invest in new machinery for that year.

Instead it may choose to buy carbon credits on the open market from organizations that have been approved as being able to sell legitimate carbon credits.

One seller might be a company that will offer to offset emissions through a project in the developing world, such as recovering methane from a swine farm to feed a power station that previously would use fossil fuel. So although the factory continues to emit gases, it would pay another group to reduce the equivalent of 20,000 tonnes of carbon dioxide emissions from the atmosphere for that year.

Another seller may have already invested in new low-emission machinery and have a surplus of allowances as a result. The factory could make up for its emissions by buying 20,000 tonnes of allowances from them. The cost of the seller's new machinery would be subsidized by the sale of allowances. Both the buyer and the seller would submit accounts for their emissions to prove that their allowances were met correctly.

2.10.2 CREATING CARBON CREDITS

The first step in determining whether or not a carbon project has legitimately led to the reduction of measurable and permanent emissions is understanding the CDM methodology process. This is the process by which project sponsors submit, through a Designated Operational Entity (DOE), their concepts for emissions reduction creation. The CDM Executive Board, with the CDM Methodology Panel and their expert advisors, review each project and decide how and if they do indeed result in reductions that are additional

Developed countries have large, binding targets and a carbon emissions quota that cannot be broken, unless they purchase carbon credits that are offset against other developed countries with binding targets on which they are below quota. The global total output is the driving consideration: if one country is below their quota, this allows another country to purchase carbon credits from them, thus giving developed countries an incentive to fall below their quota whenever possible. The Kyoto treaty has now expired and a replacement is still awaited, though international carbon crediting is still ongoing.

2.10.3 PURCHASING CARBON CREDITS

Carbon credits may be purchased on an international market where credits (and the one tonne emissions they each allow) are allocated between a group of regulated sources.

If a country or organization knows they are going to emit more carbon than their emissions quota allows over a set period, they can look to preemptively purchase carbon credits from the international market to ensure they are not penalized for their excesses.

The purchase of carbon credits can also be done reactively, with countries and organizations buying credits on the international market after a period in which they produced significantly more carbon than their forecasts predicted.

2.10.4 THE JOINT IMPLEMENTATION AND CLEAN DEVELOPMENT MECHANISMS

Purchasing carbon credits is not the only way that countries and large organizations can offset their carbon emissions in order to produce more carbon in times when doing so becomes a necessity.

Despite developed countries bearing the brunt of carbon reduction requirements, Joint Implementation encourages developed countries to work alongside developing countries to help reduce their own carbon emissions, and in turn receive carbon credits. Developing countries do not have an emissions quota, but as the goal of international protocol is to reduce carbon emissions globally, even small reductions from developing countries helps this overarching goal.

In a similar way, the Clean Development Mechanism allows developed countries to receive credits for sponsoring relevant programs in developing nations.

WHAT IS CARBON CREDIT?

In step with the dramatic rise in CO₂ emissions and other pollutants in recent years, a variety of new financial markets have emerged, offering businesses key incentives aside from taxes and other punitive measures “to slow down overall emissions growth and, ideally, global warming itself.

*A key feature of these markets is **emissions trading**, or **cap-and-trade schemes**, which allow companies to buy or sell “credits” that collectively bind all participating companies to an overall emissions limit.*

While markets operate for specific pollutants such as greenhouse gases and acid rain, by far the biggest emissions market is for carbon. In 2007, the trade market for CO₂ credits hit \$60 billion worldwide almost double the amount from 2006

How It Works

Emissions limits and trading rules vary country by country, so each emissions-trading market operates differently. For nations that have signed the Kyoto Protocol, which holds each country to its own CO₂ limit, greenhouse gas-emissions trading is mandatory. Yet a few general principles apply to each type of market.

Under a basic cap-and-trade scheme, if a company's Carbon emissions fall below a set allowance, that company can sell the difference in the form of credits to other companies that exceed their limits. Another fast-growing voluntary model is carbon offsets. In this global market, a set of middlemen companies, called offset firms, estimate a company's emissions and then act as brokers by offering opportunities to invest in carbon-reducing projects around the world. Unlike carbon trading, offsetting is not yet government regulated in most countries; it is up to buyers to verify a project's environmental worth. In theory, for every ton of CO₂ emitted, a company can buy certificates attesting that the same amount of greenhouse gas was removed from the atmosphere through renewable energy projects such as tree planting.

2.10.5 THE ADVANTAGES OF CARBON CREDITS

- Companies in different industries face dramatically different costs to lower their emissions. A market-based approach allows companies to take carbon-reducing measures that everyone can afford. The
- private sector is better at developing diversified approaches to manage the costs and risks of reducing emissions .carbon credits helps to manage the costs
- Reducing emissions and lowering energy consumption is usually good for the core business.
- There is a long-term investment angle
- Carbon credits are relatively cheap now, but their value will likely rise,

2.10.6 THE DISADVANTAGES OF CARBON CREDITS

- As with any financial market, emissions traders are vulnerable to significant risk and volatility.
- The Supply soared and carbon prices bottomed out, removing incentives for companies to trade.
- Enforcement of trading rules can be just as unpredictable.

2.11 CARBON FOOT PRINT

A carbon footprint is historically defined as **"the total sets of greenhouse gas emissions caused by an organization, event, product or person."** The total carbon footprint cannot be calculated because of the large amount of data required and the fact that carbon dioxide can be produced by natural occurrence

More practicable definition:

“A measure of the total amount of carbon dioxide (CO₂) and methane (CH₄) emissions of a defined population, system or activity, considering all relevant sources, sinks and storage within the spatial and temporal boundary of the population, system or activity of interest. Calculated as carbon dioxide equivalent (CO₂e) using the relevant 100-year global warming potential (GWP100).”

Greenhouse gases (GHGs) can be emitted through transport, land clearance, and the production and consumption of food, fuels, manufactured goods, materials, wood, roads, buildings, and services. For simplicity of reporting, it is often expressed in terms of the amount of carbon dioxide, or its equivalent of other GHGs, emitted.

Most of the carbon footprint emissions for the average household come from "indirect" sources, i.e. fuel burned to produce goods far away from the final consumer. These are distinguished from emissions which come from burning fuel directly in one's car or stove, commonly referred to as "direct" sources of the consumer's carbon footprint.

The concept name of the carbon footprint originates from ecological footprint, which estimates the number of "earths" that would theoretically be required if everyone on the planet consumed resources at the same level as the person calculating their ecological footprint..

2.11.1 MEASURING CARBON FOOTPRINTS

An individual's, nation's, or organization's carbon footprint can be measured by undertaking a GHG emissions assessment or other calculative activities denoted as carbon accounting. Once the size of a carbon footprint is known, a strategy can be devised to reduce it, e.g.

- By technological developments,
- Better process and product management,

- changed Green Public or Private Procurement (GPP), carbon capture,
- Consumption strategies,
- Carbon offsetting and others.

Several free online carbon footprint calculators exist. These websites ask you to answer more or less detailed questions about your diet, transportation choices, home size, shopping and recreational activities, usage of electricity, heating, and heavy appliances such as dryers and refrigerators, and so on. The website then estimates your carbon footprint based on your answers to these questions.

The main influences on carbon footprints include **population, economic output, and energy and carbon intensity** of the economy. These factors are the main targets of individuals and businesses in order to decrease carbon footprints. Scholars suggest the **most effective way to decrease a carbon footprint is to either decrease the amount of energy needed for production or to decrease the dependence on carbon emitting fuels.**

2.11.2 GREEN HOUSE GAS EMISSION FROM DIFFERENT MATERIALS

Food

Average greenhouse-gas emissions per day (in kilograms of carbon dioxide equivalent) were:

- 7.19 For high meat-eaters
- 5.63 For medium meat-eaters
- 4.67 For low meat-eaters
- 3.91 For fish-eaters
- 3.81 For vegetarians
- 2.89 For vegans

Textiles

Average greenhouse-gas emissions per day (in kilograms of carbon dioxide equivalent) were

- Cotton: 7
- Nylon: 5.43
- PET (e.g. synthetic fleece): 5.55
- Wool: 5.48

Cement

Cement production and carbon footprint resulting from soil sealing was 8.0 Mg person⁻¹ of total per capita CO₂ emissions (Italy, year 2003); the balance between C loss due to soil sealing and C stocked in man-made infrastructures resulted in a net loss to the atmosphere, -0.6 Mg C ha⁻¹ y⁻¹. [38]

Schemes to reduce carbon emissions: Kyoto Protocol, carbon offsetting, and certificates

Carbon dioxide emissions into the atmosphere, and the emissions of other GHGs, are often associated with the burning of fossil fuels, like natural gas, crude oil and coal. While this is harmful to the environment, carbon offsets can be purchased in an attempt to make up for these harmful effects.

2.11.3 WAYS TO REDUCE CARBON FOOTPRINT

The most common way to reduce the carbon footprint of humans is to **Reduce, Reuse, Recycle, Refuse**. In manufacturing this can be done by recycling the packing materials, by selling the obsolete inventory of one industry to the industry who is looking to buy unused items at lesser price to become competitive. Nothing should be disposed off into the soil, all the ferrous materials which are prone to degrade or oxidize with time should be sold as early as possible at reduced price.

This can also be done by using reusable items such as thermoses for daily coffee or plastic containers for water and other cold beverages rather than disposable ones. If that option isn't available, it is best to properly recycle the disposable items after use. When one household recycles at least half of their household waste, they can save 1.2 tons of carbon dioxide annually[unreliable source?].

Another easy option is to **drive less**. By walking or biking to the destination rather than driving, not only is a person going to save money on gas, but they will be burning less fuel and releasing fewer emissions into the atmosphere. However, if walking is not an option, one can look into carpooling or mass transportation options in their area.

Yet another option for reducing the carbon footprint of humans is to **use less air conditioning and heating** in the home. By adding insulation to the walls and attic of one's home, and installing weather stripping or caulking around doors and windows one can lower their heating costs more than 25 percent. Throwing food out not only adds its associated carbon emissions to a person or household's footprint, it adds the emissions of transporting the wasted food to the garbage dump and the emissions of food decomposition, mostly in the form of the highly potent greenhouse gas, methane.

2.11.4 CARBON FOOTPRINT FACTS AND STATISTICS EVERYONE SHOULD KNOW

It is time to get to the bottom of carbon footprint facts and statistics. We are often inundated with opinions and political views that skew the real facts behind our carbon footprints.

The following are some verified carbon footprint facts that can help you make an informed opinion and take advantage of opportunities to help:

Electricity = Carbon Dioxide, like it or not, every time you plug in, you are impacting the environment through carbon emissions. Power plants all over the world utilize fossil fuels, coal, and alternative energies such as nuclear, wind, water, and solar. Unfortunately there is still a carbon impact. Even discarded batteries from laptops or cell phones can leave a deadly impact in landfills. Consider unplugging for one day, or even a few hours, each week. Find an activity that does not involve electricity, and enjoy it.

HVAC and Carbon Emissions, heating and air conditioning can make up a huge spend on carbon emissions. Try to train yourself and your body to tolerate warmer temperatures in the summer months and cooler temperatures in the winter months. Wear heavier clothes in the home or wear lighter clothes depending on the climate. This will not only reduce your carbon footprint but it will also reduce your home heating and air conditioning expenses. Also, keep your HVAC units well maintained and change the air filter with the season changes.

Water Usage, the efforts to maintain and purify our drinking water can take a huge toll on carbon emissions. Consider the treatment plants and their operating equipment footprints along with all of the various chemicals needed to purify our water. Use water wisely and do not waste it.

SOURCES OF CO₂ “THE PRIMARY SOURCES OF CARBON EMISSIONS ARE

- Solid fuels such as coal 35%
- Liquid fuels such as oil or petroleum 36%
- Gaseous fuels such as natural gas 20%

Industrial gas emissions from manufacturing or processing plants - <1%

Deforestation which increases emissions by decreasing the cleaning performed by forests Paper, the more paper you use, the more you are contributing to deforestation and carbon emissions. Try switching to paperless billing as much as possible, and consider using recycled paper for sending cards, letters or wrapping gifts. If possible, use gift bags and other paper goods more than once to reduce the overall need for paper. Also unsubscribe to junk mail and catalogs that can be found online instead.

Less Driving, park your car once a day if possible and work from home or just walk to your destination. Carpooling is a great way to help reduce carbon emissions while also providing you with networking opportunities.

2.11.5 EIGHT WAYS TO REDUCE YOUR CARBON FOOTPRINT DURING THE SHORTEST MONTH THIS YEAR

1. Waste Not, Want Not

The bad news is that most people generate plenty of greenhouse gasses needlessly. The good news is that eliminating this waste does not require radical lifestyle changes. By taking a few basic steps in the major areas of your life, you can reduce your footprint almost without noticing:

Around the House: Check windows, doors, attics and basements for drafts. By sealing leaks and insulating these spaces, you can reduce your CO₂ emissions by nearly half a ton each year.

In the Shower: Everyone loves the feel of a hot shower or bath after coming in from the cold, but all that hot water takes energy to produce. Keep your showers brief and functional, and save baths for special occasions since they require twice the energy of an efficient shower. Look to install low-flow showerheads that will do the work of saving you water, energy and money for you.

In the Car: Drive consistently. Fast stops and starts eat up fuel, and while we all have places to be, driving 55 instead of 75 mph saves you between 20-30 percent in fuel efficiency.

2. Dispose of Disposables

The average household generates 4.5 pounds of waste each day, which translates to over a ton of greenhouse gas emissions from landfills each year - for one home! Therefore, the more we can divert from the waste stream, the better the carbon savings:

Use rags instead of paper towels to clean up messes. They can be reused and washed as often as you spill your milk. You can also switch to glass storage containers instead of plastic, which will eventually end up in the landfill. They may cost a bit more, but they will last for years, and you avoid the whole plastics-and-food debate.

Be sure that the items you want to trash can't be recycled. You may be surprised by just how many things people are putting to new use these days.

Use cold water for your wash and rinse cycles. Hot water accounts for 90 percent of the energy used by washers, and cold water is usually just as effective

3. Let There Be Light

With the daylight still fading by late afternoon, now is the perfect time to replace your incandescent bulbs with more energy efficient alternatives:

CFLs: Installing just 5 compact fluorescent lamps (CFLs) can reduce your CO₂ emissions by about 500 pounds per year.

LEDs: Although more expensive than CFLs, LED lights last up to five times longer. Even better, they are not affected by cold temperatures, which can limit the brightness of CFL bulbs, so they make a great long-lasting choice for outdoor lighting.

Check out the new Vu1 bulbs that will be released in the coming months. They provide the efficiency of a CFL without the mercury.

4. Taking the Out of Carbon Footprint

People are less likely to walk or use public transportation in the chilly winter months, but there are still easy ways to limit the emissions generated by your vehicle:

Carpool to work, whenever possible. Even if you catch a ride to work one day a week, thatâ€™s a 20 percent reduction in your annual work-related carbon emissions

Keep your car tuned-up regularly, including tire pressure. A well-tuned car can run up to 30 percent more efficiently, which saves both carbon and cash, which is one of many ways to drive smart

5. Donot Let That Food Go to Waste

Composting helps reduce carbon emissions in several ways. First, it keeps food, and the methane it will produce, out of the landfill. Second, organic compost reduces the need for petro-chemical fertilizers, which use fossil fuels in their refinement and distribution. You can compost a number of ways, from a pile in your backyard to a tub of worms in your basement. Either way, the carbon savings will add up quickly.

A carbon footprint is defined as

The total amount of greenhouse gases produced to directly and indirectly support human activities, usually expressed in equivalent tons of (CO₂)